

Dynamical Variational Autoencoders for audio reconstruction and generation

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Introduction

Variational Autoencoders (VAEs) are a staple in unsupervised learning for representing high-dimensional data. However, the standard VAE processes input vectors independently, making it unsuitable for sequential data where temporal dependencies are crucial.

To address this issue, the **Dynamical Variational Autoencoders (DVAEs)** framework extends VAEs by incorporating temporal dynamics into the latent space. These models combine the generative capability of deep learning with the structure of:

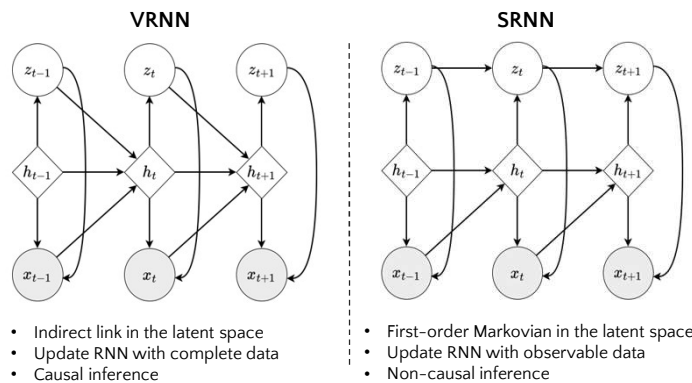
- **Recurrent Neural Networks (RNNs)** for long-term dependencies.
- **State-Space Models (SSMs)** for probabilistic handling of time.

This work specifically investigates two type of DVAEs architectures:

- **Variational RNN (VRNN)**: Fully integrates latent variables into the recurrent update, giving the internal state variability at every time step.
- **Stochastic RNN (SRNN)**: Explicitly separates the deterministic dynamics (RNN) from the stochastic latent processes (SSM), offering a structured "state-space" approach to deep generative modeling.

We investigate performance of the **VRNN** and **SRNN** with another speech dataset

VRNN and SRNN Architectures



VCTK Dataset

- It contains speech data from **109 English speakers** with diverse accents
- Approximately **400 sentences/speaker**
- We used 3000 samples of this dataset to train our models

Mel Log-Spectrogram

Time-frequency representation of sound that mimics human auditory perception.

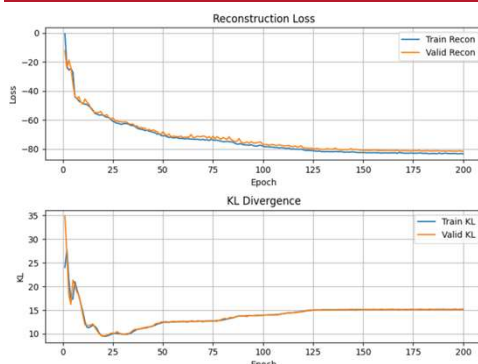
- Compute Short-Time Fourier Transform (STFT)
- Compute Mel scale:
- Transform the power into decibels (dB)

Experiments

Our objectives for this experiment were as follows:

- Re-implement the VRNN and SRNN architectures using PyTorch.
- Train the models by selecting optimal parameters and hyperparameters, focusing specifically on the choice of prior, the duration of the Kullback-Leibler (KL) divergence annealing phase, the number of epochs, and the learning rate.
- Evaluate the audio reconstruction quality across different architectures and priors (Gaussian vs. Student-t).
- Attempt to generate audio both from scratch (cold start) and using a warm-up phase (priming).
- Refine the generated audio and the models to achieve more intelligible results.

Training models



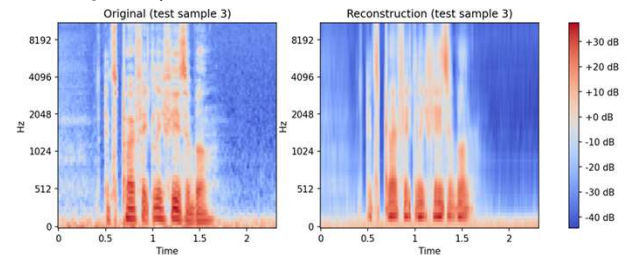
Satisfying training curve:

- Reconstruction loss smoothly decreases with epochs.
- KL decreases sharply during annealing phase, then increases slowly, meaning that our model properly exploits the latent space.

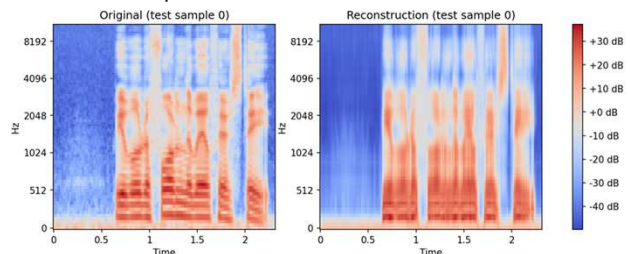
Audio reconstruction

Regarding audio reconstruction, the VRNN with Gaussian prior and the VRNN with Student-t prior (df=3) yield similar results and the SRNN slightly underperforms.

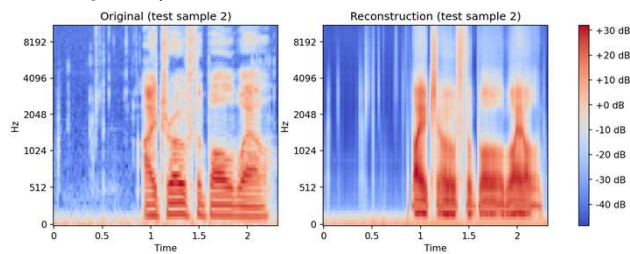
VRNN with gaussian prior:



VRNN with Student t prior:

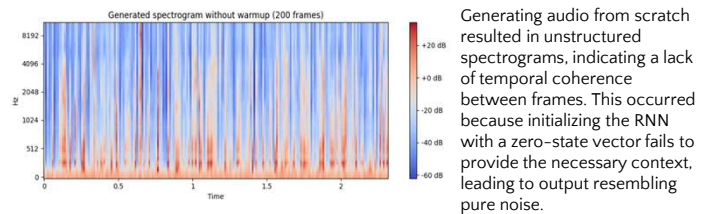


SRNN with gaussian prior:



Limitation of this approach: audio is reconstructed from mel-spectrograms without phase information, resulting in robotic voices. The lack of phase data masks fine acoustic details (timbre, pitch,...) making it difficult to precisely assess the quality of the reconstruction.

Audio generation



To address this, we implemented a warm-up (priming) phase to stabilize the model's hidden state with real data, and we applied a Gaussian smoothing filter to remove sharp spectral discontinuities. We also adjusted the latent variable variance.

Despite these optimization attempts, improvements were minimal; the audio remained unintelligible, sounding either like underwater speech or static from an untuned radio, depending on the variance settings.

Conclusion

- VRNN and SRNN effectively compress data into a low-dimensional latent space, achieving satisfactory reconstruction even with a small subset of the VCTK dataset.
- However, generation quality remains poor; the RNN struggles with autoregressive prediction, resulting in unstructured spectrograms.
- Future work should involve training on the full dataset and exploring other architectures such as RVAE, DSAE, STORN, or generative paradigms like Diffusion Models and GANs.

References

Girin, Laurent, et al. "Dynamical variational autoencoders: A comprehensive review."